

Andrew G. Benson, Energy Economist

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Energy economist and strategist with expertise in nuclear energy, energy storage, electricity grids, and energy policy. Proven ability to analyze complex technical issues, build models, integrate knowledge across domains, and present results clearly to support executive decision-making.

Core Competencies

Applied Microeconomics
Industrial Organization
Political Economy

Energy Economics
Techno-Economic Assessment
Corporate Strategy

Forecasting
Casual Inference
Database Design

Professional Experience

Kairos Power

Senior Economist

2025 [May] – Present

Senior Strategy Analyst

2023 [Jan] – 2025 [May]

- Worked as an individual contributor within the Strategy Team, providing cross-functional analytical support to engineering, finance, government affairs, and executive leadership teams
- Designed and maintained a database, user interface, and dashboard for tracking progress in manufacturing, assembly, and commissioning of \$26 million hardware test
- Supported the conceptual development and negotiation of novel commercial arrangements with Kairos Power's lead customer to enable a power-purchase offtake agreement for new nuclear generating assets
- Integrated new data sources into my global dataset of commercial nuclear power plants (originally assembled during my PhD) to estimate techno-economic parameters for financial models
- Analyzed internal procurement data to generate probabilistic forecasts of purchase order lead time; identified characteristics of purchase orders associated with long-lead times and missed delivery dates
- Lead analyst for production cost modeling of FLiBe salt and power plant operations & maintenance

Sandia National Laboratories

Energy Storage Technology and Systems Department

Postdoctoral Appointee

2021 [Aug] – 2023 [Jan]

- Developed model to predict installed cost of behind-the-meter battery energy systems, with estimates customized to technical specifications of individual projects and empirically calibrated uncertainty
- Co-authored research on energy storage policy, renewables integration, and grid expansion planning
- Analyzed FERC Form 1 data to estimate historical CAPEX and OPEX for pumped storage hydropower

University of California, Irvine

Teaching Assistant

2016 [Sep] – 2021 [Jun]

- Supported instruction, assignment design, and grading for courses in economics and political science
- Teaching effectiveness rated "adequate" or better by 90% of students, and exceeding "adequate" by 66% of students (N = 476)

California Energy Commission

Energy Research & Development Division

Associate Energy Specialist

Energy Analyst

Student Assistant

2015 [Jan] – 2016 [Jul]

2013 [Sep] – 2014 [Dec]

2012 [Jul] – 2013 [Aug]

- Performed retrospective cost-benefit evaluation of R&D grants
- Contributed to annual reports to the state Legislature on program outcomes
- Supported development of R&D program budgets
- Supervised and mentored three interns

Education

University of California, Irvine

Ph.D., Economics

2021

M.A., Economics

2017

University of California, Davis

B.A., Economics and Political Science

2013

Technical Skills

Programming Languages: Python, VBA, SQL, Stata, L^AT_EX

Software: Microsoft Excel, Tableau

Selected Publications

Customized Predictions of the Installed Cost of Behind-the-Meter Battery Energy Storage Systems, in *Energy Reports* (2023). <https://doi.org/10.1016/j.egyr.2023.04.270>

Long-Duration Energy Storage in a Decarbonized Future: Policy Gaps, Needs, and Opportunities, jointly with Will McNamara, Valerio DeAngelis, Ray Byrne, Babu Chalamala, and Ralph Masiello, in *MRS Energy & Sustainability* (2022). <https://doi.org/10.1557/s43581-022-00037-9>

An Analysis of PNM's Renewable Reserve Requirements to Meet New Mexico's Decarbonization Goals, jointly with Jim Ellison and Cody Newlun (2022). SAND Report 2022-6704. <https://www.osti.gov/servlets/purl/1868430/>

Global Divergence in Nuclear Power Plant Construction: the role of political decentralization, in *Nuclear Technology* (2022). <https://doi.org/10.1080/00295450.2021.1991762>